

Bahram Behzadian

Reinforcement Learning | Post-Training | Evaluation

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Profile

Research scientist with a Ph.D. in reinforcement learning studying policy improvement when the model or evaluator is imperfect. My work spans robust MDPs under model uncertainty, recurrent value evaluators with verifier-style feedback, and production experimentation systems for measurement correctness and reliability. Current interests include RL post-training, reasoning, long-horizon agents, and reliable tool-use decisions under uncertainty.

Experience

2022–Present **Research Scientist**, *Meta*

- Build production systems for experiment-metric correctness and ML auditing in large-scale ads experimentation.
- Develop anomaly and data-quality monitoring together with statistical diagnostics for unreliable, biased, stale, or inconsistent signals.
- Translate failure analyses into monitoring and correctness safeguards for production experimentation systems.

2018 **AI Engineer Intern**, *Envio, Inc.*, Dover, NH

- Developed learning-based methods for vehicle routing and scheduling under operational constraints and uncertain execution.

Selected Research

Policy Improvement with Imperfect Evaluators

Finite-Reference Error Bounds for Recurrent Value Evaluators in Policy Improvement (UPI-TRM); earlier version at ICLR 2026 Workshop on AI with Recursive Self-Improvement

Co-developed finite-depth error bounds for recurrent value evaluators used in policy improvement without assuming recurrent convergence, separating endpoint discrepancy from Bellman-residual error. Connected centered advantage error to conservative policy-improvement bounds and evaluated the analysis on an exact finite MDP and a fixed Sudoku census.

Research-Scale Language-Model Post-Training

Implemented PPO and DPO experiments on small language models for arithmetic tasks with exact-answer evaluation.

Robust and Risk-Sensitive Markov Decision Processes

NeurIPS 2021 and AISTATS 2021

Developed scalable algorithms for L_∞ -constrained S-rectangular robust MDPs and percentile criteria under model uncertainty.

Technical Skills

Programming: Python, PyTorch.

Evaluation and experimentation: experiment-metric correctness, statistical diagnostics, anomaly and data-quality monitoring, ML auditing.

RL and post-training: robust MDPs, policy improvement, research-scale PPO and DPO, verifier-style rewards and RL with verifiable rewards; working knowledge of GRPO and reward modeling.

Selected Publications

UPI-TRM: Policy Improvement under Truncated Evaluators with Verifier-Style Rewards

Bahram Behzadian, Brett Daley, Gopeshh Subbaraj, Houssam Nassif

ICLR 2026 Workshop on AI with Recursive Self-Improvement [Paper]

Fast Algorithms for L_∞ -Constrained S-Rectangular Robust MDPs

Bahram Behzadian, Marek Petrik, Chin Pang Ho

Conference on Neural Information Processing Systems (NeurIPS), 2021 [Paper]

Optimizing Percentile Criterion using Robust MDPs

Bahram Behzadian, Reazul Hasan Russel, Marek Petrik, Chin Pang Ho

International Conference on Artificial Intelligence and Statistics (AISTATS), 2021 [Paper]

Fast Feature Selection for Linear Value Function Approximation

Bahram Behzadian, Soheil Gharatappeh, Marek Petrik

International Conference on Automated Planning and Scheduling (ICAPS), 2019 [Paper]

Monte Carlo Localization in Hand-Drawn Maps

Bahram Behzadian, Pratik Agarwal, Wolfram Burgard, Gian Diego Tipaldi

IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2015 [Paper]

Complete publication list: [Publications page](#) | [Google Scholar](#)

Education

2015–2022 **Ph.D. in Computer Science**, *University of New Hampshire*

Dissertation: *Efficient Data-Driven Robust Policies for Reinforcement Learning* (advisor: Marek Petrik).

M.S. in Computer Science awarded en route (2018). Thesis: *Representation Learning for Reinforcement Learning via Singular Value Decomposition*.

2010–2013 **M.S. in Machine Automation**, *Tampere University*

Thesis: *Robot Localization under Partial Observability with Weak Maps*

Honors and Fellowships

CEPS Graduate Fellowship, University of New Hampshire

Thesis and Dissertation Fellowship, Tampere University of Technology